

Quaternion Algebras

Preliminary Arizona Winter School, 2026

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Introduction

Welcome!

These notes about quaternion algebras are written for the Preliminary Arizona Winter School in 2026 and aimed at an audience of undergraduates with a first course in abstract algebra. If that does not describe you, then that doesn't mean these notes aren't for you! However, you may want to brush up on a few concepts from linear and abstract algebra including bases, dimension, ideals, ring homomorphisms, field extensions, and finite fields. Algebra texts such as [DF04, Chapters 7,8] or [Sil22, Chapters 3–5] are a great place to start.

The notes are meant to be self contained, but I hope that they will spark your curiosity and you will feel inspired to explore this beautiful subject more deeply. Some textbook references that you might find helpful as a starting point include [GS17, Ser73, Voi21], although all of these are at the graduate level. Another source freely available online is [Conb] which was also invaluable to me while writing these notes.

Scope

Quaternion algebras are a vast subject. These notes take a somewhat narrow perspective with an eye towards the topic of the 2026 Arizona Winter School: Rationality and Irrationality: Arithmetic and Geometry. These notes introduce quaternion algebras from an algebraic perspective with a goal of distinguishing between quaternion algebras over the rational numbers via arithmetic and geometry. Five lectures is short! There are many aspects of quaternion algebras that we must skip. One that may be of interest to the casual reader is their connections to visualizations; see for example [Han06, Chapters 1 and 2].

What will we cover?

Here is a brief roadmap of what we'll cover in the five lectures:

- **Lecture 1 (Hamilton quaternions and foundations):** We start with Hamilton’s quaternions $\mathbb{H}$ as historical motivation before generalizing to arbitrary quaternion algebras $(a, b)_F$ over a base field F .
- **Lecture 2 (Isomorphisms, bases, and splitting):** We characterize when two quaternion algebras are isomorphic and explore the concept of *splitting*—determining when $(a, b)_F \cong M_2(F)$ versus when it remains a division algebra.
- **Lecture 3 (Central simple algebras & associated conics):** We place quaternion algebras into the broader context of central simple algebras (CSAs) and establish a connection between algebra and geometry by associating a conic curve to each quaternion algebra.
- **Lecture 4 (A primer on p -adics):** We introduce the p -adic integers $\mathbb{Z}_p$, p -adic valuations to give us the machinery needed to understand behavior of quaternion algebras over the rational numbers in the final lecture.
- **Lecture 5 (Quaternion algebras over $\mathbb{Q}$):** We define a tool called the Hilbert symbol $(a, b)_p$ and apply a local-global principle to distinguish non-isomorphic quaternion algebras over $\mathbb{Q}$.

How should you use these notes?

As many others have probably said before me, mathematics is not a spectator sport! Even reading mathematics should be active rather than passive. When you encounter a new definition, try working through an example. Challenge yourself to come up with your own example or instructive non-example. Additionally, you should make some time to work through the inline exercises scattered throughout the text (which are distinct from the problem sets accompanying the course).

What if you find a typo?

I hope there won’t be many, but inevitably there will be some. Just send me an email. I appreciate it!

Acknowledgements

This is a draft. Add more stuff here after people have actually read the notes and given feedback. Thank the project leaders for their hard work and the PAWS organizers. :)

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Lecture 1: Hamilton quaternions and foundations

1.1 Motivation

The construction of the complex numbers $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ provides a bridge between algebra and geometry. In the 2-dimensional plane, addition of complex numbers corresponds to vector addition or translation, while multiplication encodes scaling and rotations. For instance, multiplying a vector by i corresponds to a 90° rotation.

Given the correspondence between planar rotations and multiplication in the complex numbers, a natural question arises. Can we construct a similar 3-dimensional algebra to model rotations in 3-dimensional space? William Rowan Hamilton spent over a decade searching for such a system (from roughly 1830 until his breakthrough on October 16, 1843). He eventually realized that a 3-dimensional associative division algebra over $\mathbb{R}$ is an algebraic impossibility. To preserve the ability to divide by non-zero elements while extending the complex numbers, we must make a structural compromise. By expanding the construction to a 4-dimensional algebra and dropping the axiom of commutativity, Hamilton arrived at the quaternions, denoted $\mathbb{H}$. This algebra serves as our archetypal example of a non-commutative division ring and forms the foundation for the general theory of quaternion algebras we will develop in this course.

1.2 Hamilton's quaternions

Definition 1.1: Hamilton Quaternions

The **(Hamilton) quaternions** are

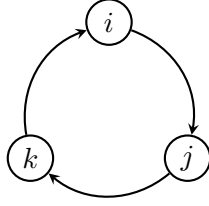
$$\mathbb{H} := \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\},$$

with multiplication defined by the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k = -ji, \quad \text{and every } a \in \mathbb{R} \text{ commutes with } i, j, k.$$

Just as the complex numbers form a 2-dimensional $\mathbb{R}$ -vector space with basis $\{1, i\}$, the Hamilton quaternions form a 4-dimensional $\mathbb{R}$ -vector space spanned by $\{1, i, j, k\}$. Equipped with the multiplication rules above, $\mathbb{H}$ becomes an associative ring. A helpful way to remember the rules

for multiplication in $\mathbb{H}$ is to arrange i, j, k clockwise in a circle. Moving along the arrows gives a positive product (e.g., $ij = k$, $jk = i$, $ki = j$), while moving against the arrows gives a negative product (e.g., $ji = -k$, $kj = -i$, $ik = -j$).



The rules for multiplication among i, j, k together with the distributive law are all we need to multiply any elements in $\mathbb{H}$.

Example 1.2. $(i + 2j)(2i - j) = 2i^2 - ij + 4ji - 2j^2 = -2 - k - 4k - (-2) = -5k$.

Before proving structural properties, it is helpful to get some hands-on practice calculating directly with elements in $\mathbb{H}$.

Exercise 1.3. Let $q = 1 + 2i - j$ and $r = 2i + k$ be quaternions in $\mathbb{H}$.

- (i) Compute the products qr and rq . Verify explicitly that $qr \neq rq$.
- (ii) Calculate q^2 and r^2 .
- (iii) Compute $(q + r)(q - r)$ and compare it to $q^2 - r^2$. Why are these two expressions not equal in $\mathbb{H}$?

To get a better sense of how non-commutativity controls arithmetic in $\mathbb{H}$, we can ask: which quaternions still commute with every element, and how many square roots of -1 exist?

Exercise 1.4.

- (i) The **center** of $\mathbb{H}$ is defined as $Z(\mathbb{H}) = \{q \in \mathbb{H} \mid qx = xq \text{ for all } x \in \mathbb{H}\}$. Prove that $Z(\mathbb{H}) = \mathbb{R}$.
(Hint: Write $q = a + bi + cj + dk$ and compute $qi - iq$ and $qj - jq$.)
- (ii) Find all **pure quaternions**, i.e., elements of the form $q = bi + cj + dk$, that satisfy $q^2 = -1$. How does your result compare with the complex numbers $\mathbb{C}$?

The Hamilton quaternions $\mathbb{H}$ are built over the real numbers with $i^2 = j^2 = -1$. However, we can abstract this construction by replacing $\mathbb{R}$ with an arbitrary base field F , and allowing i^2 and j^2 to equal any non-zero scalars $a, b \in F^\times$. While $\mathbb{H}$ is optimized for geometry over $\mathbb{R}$, extending

this framework to other fields including the rational numbers or finite fields turns out to give us a powerful tool in number theory.

In these settings, quaternion algebras provide arithmetic machinery used to study quadratic forms, prime factorization, and the classification of non-commutative rings. This leads us directly to our general definition.

1.3 General quaternion algebras

From now on, F is assumed to be field with characteristic not equal to 2.

Definition 1.5: Quaternion algebra

Let F be a field and let $a, b \in F^\times$. The **quaternion algebra** $(a, b)_F$ is the 4-dimensional F -algebra with basis $\{1, i, j, ij\}$ and multiplication determined by the relations

$$i^2 = a, \quad j^2 = b, \quad ij = -ji.$$

Remark 1.6. Let's clarify two pieces of terminology we'll use throughout these notes:

- An **F -algebra** A is both an F -vector space and a ring, where scalar multiplication and ring multiplication are compatible, i.e.,

$$(cx)(dy) = (cd)(xy) \quad \text{for all } c, d \in F \text{ and } x, y \in A.$$

- A **division algebra over F** is an F -algebra D in which every non-zero element $v \in D$ has a multiplicative inverse $v^{-1} \in D$.

Note: A division algebra is simply a field if multiplication is commutative, or a non-commutative “skew field” like $\mathbb{H}$ if it is not.

Aside 1.7. Why do we enforce $\text{char}(F) \neq 2$? Recall that in a field of characteristic 2, we have $1 + 1 = 0$, which implies $1 = -1$. If we allowed $\text{char}(F) = 2$, our core non-commutative relation $ij = -ji$ would become $ij = ji$. The algebra would become a commutative ring! It is worth noting that quaternion algebras do exist in characteristic 2, but they require a completely different set of structural relations (namely, $i^2 + i = a$, $j^2 = b$, and $ji = (i + 1)j$) to preserve non-commutativity. We will leave the characteristic 2 setting aside for this course, but the interested reader should consult [GS17, Section 1.2] for further details.

Example 1.8 (Quaternion algebras over $\mathbb{Q}$). *By varying the parameters $a, b \in \mathbb{Q}^\times$, we obtain different 4-dimensional algebras over the rational numbers:*

- In $(-1, -1)_{\mathbb{Q}}$, we have $i^2 = -1$ and $j^2 = -1$. This is simply Hamilton's quaternions restricted to rational coordinates $a, b, c, d \in \mathbb{Q}$.
- In $(2, 3)_{\mathbb{Q}}$, we have $i^2 = 2$, $j^2 = 3$, and $ij = -ji$. An element looks like $v = v_0 + v_1i + v_2j + v_3ij$ with $v_k \in \mathbb{Q}$, where $ij \cdot ij = (ij)(-ji) = -i(jj)i = -i(3)i = -3i^2 = -6$.

To study the structure of general quaternion algebras, we need an algebraic mechanism to extract scalar quantities from quaternion elements. Recall that for a complex number $z = x + iy$, multiplying by its complex conjugate $\bar{z} = x - iy$ yields a real number, $|z|^2 = x^2 + y^2$, which measures its squared magnitude. We can generalize this construction to any quaternion algebra $(a, b)_F$ by defining a conjugation map that allows us to recover an element of the base field F via multiplication, giving us an algebraic analogue of squared magnitude known as the *norm*.

Definition 1.9: Conjugate and Norm

Let $v = v_0 + v_1i + v_2j + v_3ij$ be an element of the quaternion algebra $(a, b)_F$. The **conjugate** of v is

$$\bar{v} := v_0 - v_1i - v_2j - v_3ij$$

and the **norm** of v is

$$N(v) := v\bar{v} = v_0^2 - av_1^2 - bv_2^2 + abv_3^2.$$

The map $(a, b)_F \rightarrow (a, b)_F$ defined by $v \mapsto \bar{v}$ is called an **anti-automorphism** of the F -algebra $(a, b)_F$, which means it is an F -vector space isomorphism of $(a, b)_F$ satisfying $vw \mapsto \overline{vw} = \bar{w}\bar{v}$. Moreover $\bar{\bar{v}} = v$, and an anti-automorphism with this property is called an **involution** in ring theory. For more information about involutions and central simple algebras, see [KMRT98].

Proposition 1.10: Invertible elements in quaternion algebras

An element q of the quaternion algebra $(a, b)_F$ is invertible if and only if it has nonzero norm. In particular, $(a, b)_F$ is a division algebra if and only if the norm $N: (a, b)_F \rightarrow F$ does not vanish except at 0.

Proof. The norm is a multiplicative function since

$$N(q_1q_2) = q_1q_2\overline{(q_1q_2)} = q_1q_2\bar{q_2}\bar{q_1} = q_1N(q_2)\bar{q_1} = q_1\bar{q_1}N(q_2) = N(q_1)N(q_2).$$

(Since $N(q_2) \in F$ is a scalar in the base field, it commutes with the quaternion $\overline{q_1}$.)

If $qq' = 1$ for some element q' then $N(q)N(q') = N(1) = 1$ in F thus $N(q) \neq 0$. Conversely, if $N(q) \neq 0$, the quaternion $\overline{q}/N(q)$ is a two-sided multiplicative inverse for q . $\square$

Proposition 1.10 provides a straightforward algebraic criterion for invertibility. To see this in action, we can ask whether the norm can actually vanish for a non-zero element, or if every quaternion algebra is automatically a division algebra. The following example demonstrates that the norm can indeed vanish, even over familiar fields like the rational numbers.

Example 1.11. Consider the quaternion algebra $A = (1, -1)_{\mathbb{Q}}$, where $i^2 = 1$ and $j^2 = -1$. Let's examine the element $v = 1 + i$. Its norm is

$$N(1 + i) = (1 + i)(1 - i) = 1 - i + i - i^2 = 1 - 1 = 0.$$

Since $N(v) = 0$ but $v \neq 0$, Proposition 1.10 tells us that $1 + i$ is not invertible. In fact, it is a zero divisor! Thus, $(1, -1)_{\mathbb{Q}}$ is an example of a quaternion algebra that is not a division algebra.

Exercise 1.12. For each of the following quaternion algebras, determine whether it is a division algebra or not. Justify your answer either by demonstrating that the norm map has no non-trivial zeros or by explicitly constructing a non-zero zero divisor.

- (i) $A_1 = (-1, -1)_{\mathbb{R}}$ (The classical Hamilton quaternions $\mathbb{H}$)
- (ii) $A_2 = (2, 2)_{\mathbb{Q}}$
- (iii) $A_3 = (1, 5)_{\mathbb{Q}}$
- (iv) Come up with your own examples!

Hint: For (i), inspect the signs of the coefficients in the norm form. For (ii) and (iii), write out the norm form and look for small integer choices for v_0, v_1, v_2, v_3 that cause the terms to cancel out to zero.

Looking Ahead

In this introductory lecture, we have established the foundational structure of quaternion algebras and observed that their algebraic structure is closely tied to the behavior of the quadratic norm form $N(v)$. Specifically, Proposition 1.10 shows that whether an algebra admits a division ring structure depends entirely on whether the equation $N(v) = 0$ has non-trivial solutions.

In the next lecture, we will show that the presence of these non-zero zero divisors implies that the algebra is isomorphic to the matrix ring $M_2(F)$, a property known as being *split*. This sets up the central theme of the course: classifying quaternion algebras up to isomorphism over various base fields. We will achieve this by translating our algebraic questions into a geometric setting, where finding zero divisors corresponds to finding solutions to an equation that defines a curve. From there, we will explore the behavior of quaternion algebras over finite fields, and eventually introduce specialized number systems called p -adic fields that allow us to study arithmetic one prime number at a time. Ultimately, we will see how combining this “localized” information allows us to determine the behavior of a quaternion algebra over the rational numbers.

Key Takeaways

Non-commutativity: Extending the complex numbers to a 3-dimensional associative division algebra over $\mathbb{R}$ is not possible. To build a higher-dimensional analogue of $\mathbb{C}$ with these properties, we must expand to 4 dimensions and drop commutativity ($ij = -ji$).

Algebra is governed by arithmetic: A general quaternion algebra $(a, b)_F$ is a division ring if and only if its associated norm form $N(v) = v_0^2 - av_1^2 - bv_2^2 + abv_3^2$ has no non-trivial zeros.

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